

Course «C Programming Language»

Topic: Preprocessor

Homework

Exercise 1.

Create a project, in which:

1. Create three files: (function.h, function.cpp, prog.cpp).
2. In a file function.cpp you should write the following functions to operate with an array of data:
 - a) function to fill the array with random values;
 - b) function to output values of the array to the console;
 - c) function to search for minimum element;
 - d) function to search for maximum element;
 - e) sorting function;
 - f) function to edit value of the array.

These functions must be written to operate with an array of integers, real and symbolic values. (For each type you should write separate function).

3. In a file function.h you should:
 - a) describe prototypes of all functions;
 - b) check whether constant, which indicates a type of data, is defined (`#ifdef INTEGER`);
 - c) create generalizing function names, which will get a function of the appropriate type, depending on the constant specified in prog.cpp file. (`# Define show ShowInt`);
4. In the file prog.cpp you should:
 - a) in main function call all functions through generalized names of function.h file (`show For ( )`);
 - b) define a constant that indicates what type of data will be used. For example, `INTEGER` - whole data, `CHAR` - character data, `DOUBLE` - real data (`# define INTEGER`).

Exercise 2.

Write the following macros using `#define` directive:

- find the smaller of two numbers;
- find the larger of two numbers;
- raise a number to the second power;
- raise a number to a power;
- check whether a given number is even;
- check whether a given number is odd.